Principles of movement augmentation

JONATHAN EDEN

Human movement augmentation

- Augmentation: to extend a person's movement abilities

- Unlike prosthetics or rehabilitation, augmentation does not need to substitute a lost ability with the same function, and is not bounded to a natural appearance

Human movement augmentation

- Augmentation extends the body's natural abilities

Human movement augmentation

- Augmentation extends the body's natural abilities
  - Facilitated by technology
  - Ideally naturally integrates into daily activities

Types of augmentation

- Augmentation can alter the **sensing** or **motor** abilities of users

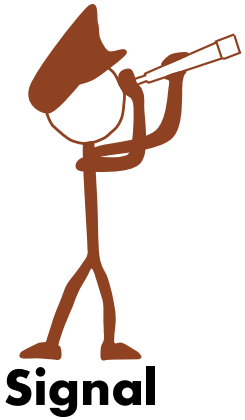
Sensory Augmentation

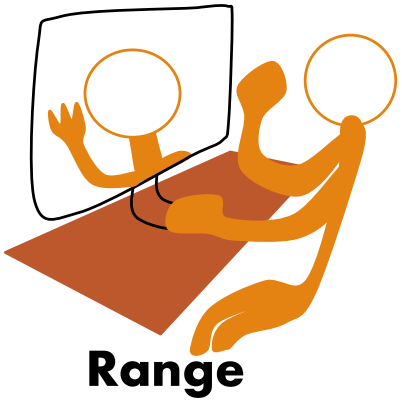
Motor Augmentation

Sensory augmentation

Sensory augmentation

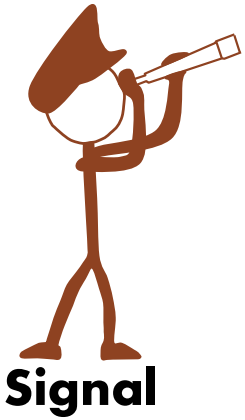
- **Signal augmentation**
- **Range augmentation**
- **Channel augmentation**





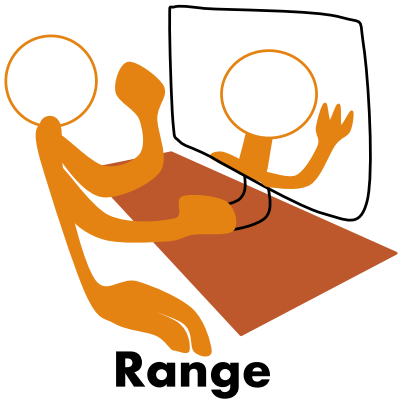
Sensory augmentation

- **Signal augmentation** alters the signal characteristics (amplitude or frequency spectrum) before it is received by the body
  - e.g. telescope, hearing aid, night vision-goggles



Sensory augmentation

- **Range augmentation** extends the sensing range by integrating additional sensors
  - e.g. zoom call, third eye



Sensory augmentation

- **Technology:** Third eye
- **Application:** Avoiding obstacles while looking at your phone

[https://www.dezeen.com/2021/05/06/minwook-paeng-third-eye-design/]

Sensory augmentation

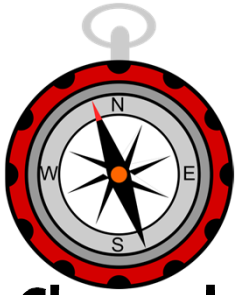
Sensory augmentation

- **Technology:** Third eye
- **Application:** Avoiding obstacles while looking at your phone
- **Requirements:** camera, wireless connection, infrared sensor, IMUs and a phone app

[https://www.dezeen.com/2021/05/06/minwook-paeng-third-eye-design/]

Sensory augmentation

- **Channel augmentation** extends the possible augmentation channels that can be sensed
  - e.g. compass, GPS



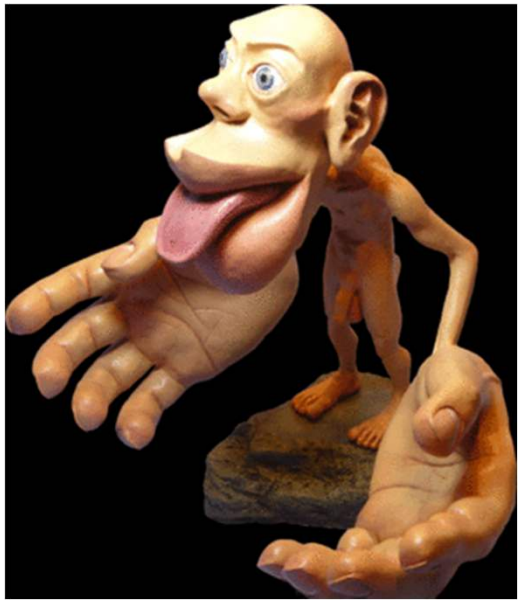
Channel

Sensory augmentation

- **Technology:** Oxygen level monitor
- **Application:** identifying a loss of oxygen and calling for assistance
- **Requirements:** photo transmitter, photodetector, wireless connection

Sensory augmentation

- Human sensing is a finite resource
  - Natural sensing resolution differs across the body
- Sensory augmentation is not free
  - There will be interference!

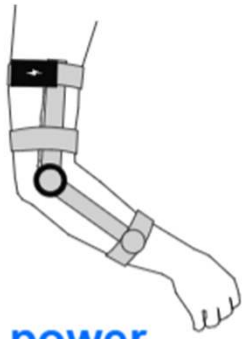


Motor augmentation

Motor augmentation

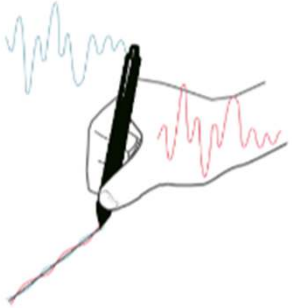
- **Power augmentation**
- **Workspace augmentation**
- **Precision augmentation**
- **DoF augmentation**

DoF



power

workspace

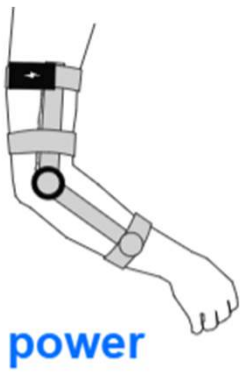


precision

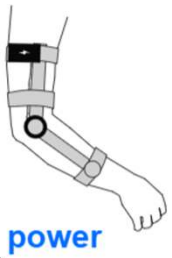
[Eden et al. 2022 Nature Communications]

Motor augmentation

- **Power augmentation** increases the users speed or force
  - e.g. levers, bicycles, exoskeletons



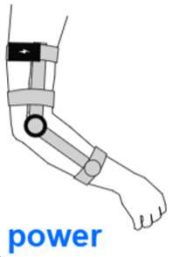
Motor augmentation



- **Technology:** Exosuit
- **Application:** Reducing load on a user during walking
- **Requirements:** Cables, harness, motors

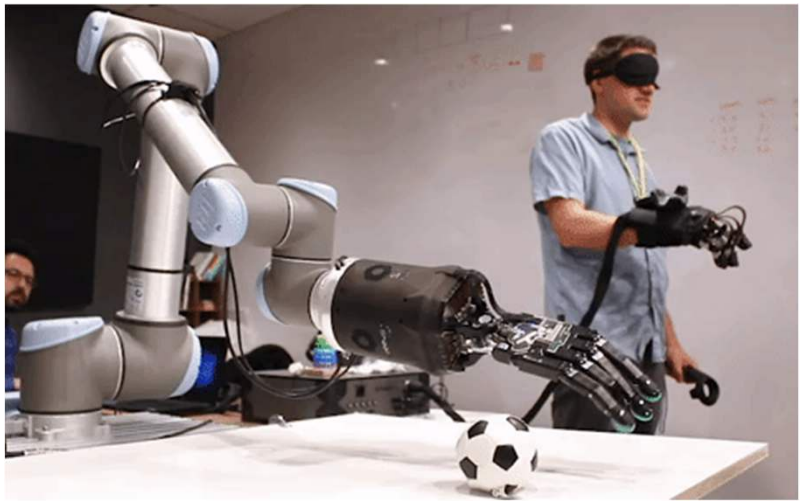
[https://biodesign.seas.harvard.edu/]

Motor augmentation



Motor augmentation

- **Workspace augmentation** extends the spatial reach of the individual
  - e.g. rake, telerobotics

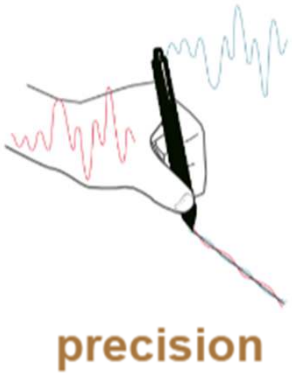


Motor augmentation

- **Technology:** Robotic surgery
- **Application:** Multi-tool surgery at a distance
- **Requirements:** Haptic interfaces, robotic arms, wireless connection, cameras

Motor augmentation

- **Precision augmentation** increases the precision/accuracy during motor actions
  - e.g. active noise cancellation, gait stabilisation devices



Motor augmentation

precision

- **Technology:** Tremor cancelation device
- **Application:** To stabilise tools for people with Parkinson's disease
- **Requirements:** IMUs, onboard MCU, 2 motors

[www.google.com/liftware]

Motor augmentation

Motor augmentation

- **DoF augmentation** extends the effective number of degrees of freedom of the user
  - e.g. supernumerary legs, fingers and arms

DoF

[Yang et al. 2022 IROS]

[Véronneau et al. 2020 Robotics and Automation Letters]

Motor augmentation

- **Technology:** Third thumb
- **Application:** Extra grasp and manipulation assistance
- **Requirements:** 2 motors, 2 pressure sensors, wireless transmitter

[Kleiba et al. 2021 Science Robotics]

Motor augmentation

DoF

Robotic hand augmentation drives changes in neural body representation

Paulina Kieliba, Danielle Clode, Roni O. Maimon-Mor, Tamar R. Makin

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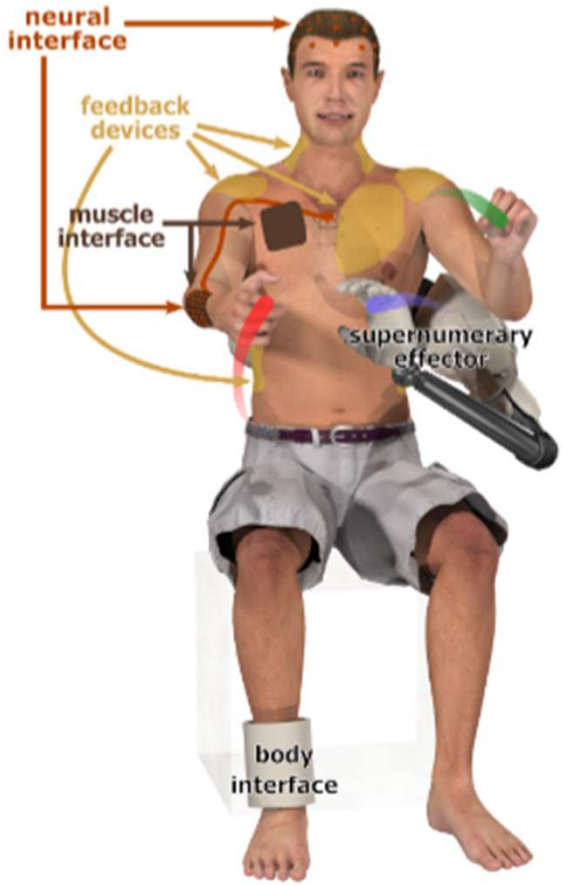
Components of augmentation

Command interface

The interface that converts user intention into commands

Feedback device

The device that gives the user status knowledge



Actuator

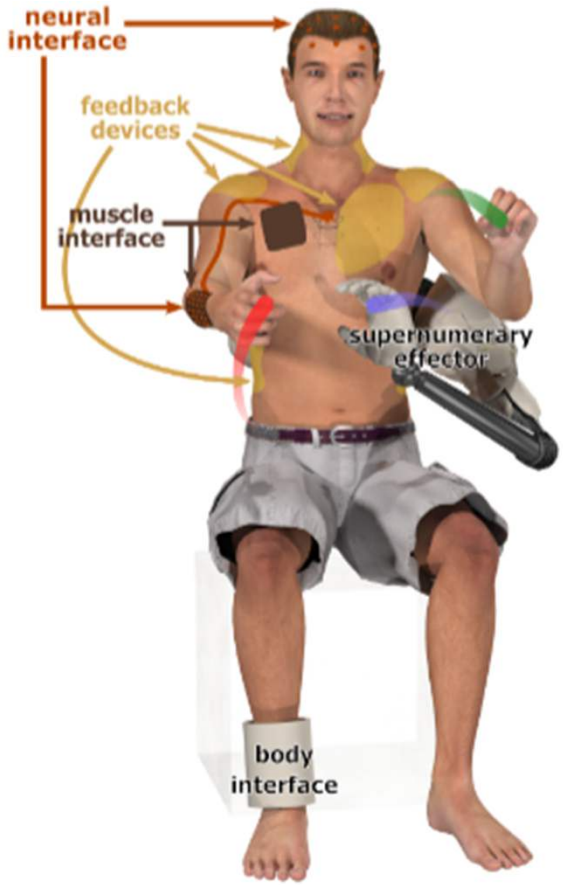
The device/s that allows the user to create motion

[Eden et al. 2022 Nature Communications]

Components of augmentation

Command interface
The interface that converts user intention into commands

Feedback device
The device that gives the user status knowledge



Actuator
The device/s that allows the user to create motion

+ **software** to control it and link it all together

[Eden et al. 2022 Nature Communications]

Thank you!